AND

LEFT	RIGHT	RESULT
0	0	0
1	0	0
1	1	1

OR

LEFT	RIGHT	RESULT
0	0	0
1	0	1
1	1	1

NOT

LEFT	RESULT	
0	1	
1	0	

© 2026 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI.

Control Structures

1. Deciding between two or more paths
2. Repeating certain steps multiple times
3. Stop doing something when a condition is met

Control structures help in making code:
1. simple - no repeating code
2. smarter - it can react
3. Flexible - it can perform multiple things depending on the situation

1. Decision Making (Conditional Statements)
2. Repetition (Loops)
3. Branching or Jumps (Switching between or skip steps)

1. Decision Making

IF - ELSE IF - ELSE

2. Loops
Running the same operation until a condition is met.

- while loop
- for loop
- do while loop

© 2026 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI.

while loop can cause infinite executions if conditional expression is not controlled.

3. Branching / Jumps
- Switch
- Functions - structured jumps and reusability

What is a Function?
Function is a named block of code that can:
- perform a specific task
- can be reused
- can take inputs
- can return an output
- can be called anywhere in the program

Why Function is called a control strucutre?
- Calling a function jumps to that code
- after execution, control returns to where it was called
- they can be nested, conditional or recursive

```
function makePizza(toppings) {
    prepareDough()
    addToppings(toppings)
    bake()
    return 'Pizza is ready'
}

let response = makePizza(['capsicum','onion'])
console.log(response)
```

© 2026 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI.

Objects

```
const person= {
    name: 'Alice',
    city: 'New Delhi'
    age: 26
}
```

for ... in loop to iterate through each key and get the value
Object.keys() to get the list of all the available keys
Object.values() to get the values of all the keys present in list
Object.entries() - creates a list of an array of key value pairs

Arrays
let fruits = ['apple', 'banana', 'mango']

using .length property we can get the number of elements present in array

.forEach()
- executes a function each time
- it does not return anything

.map()
- returning a new array
- iterates all the elements keeping the initial array as it is

.filter()
- returns a new array
- filter out the data as per the condition provided

.reduce()
- Accumulates values into a single result
- can be used to calculate averages or totals

© 2026 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI.